

Hassan's Restaurant Mobile App

UML Diagrams & Functional Overview

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Functional Overview

Hassan's Restaurant Mobile App is a React Native (Expo) application that supports two user roles — Customer and Manager — each with a dedicated set of screens and permissions. The sections below summarize what each part of the app does.

Authentication

- Login and Signup on a single screen, toggled by the user.
- Form validation for email format and password strength.
- Role selection (Customer or Manager) at signup, controlling which part of the app the user can access.
- Simulated network delay with a loading indicator to mimic a real backend call.

Menu Browsing (Customer)

- Menu items load with a simulated fetch delay, with pull-to-refresh support.
- Filter by category (Starters, Mains, Desserts, Drinks).
- Debounced live search across item name and description.
- Sort by price (low-high / high-low) or name.
- Mark items as favorites; unavailable items are shown greyed out and cannot be ordered.

Cart & Checkout (Customer)

- Add, remove, increment, or decrement items in the cart.
- Add special instructions per item (e.g. "no onions").
- Apply promo codes for a percentage discount.
- Order Summary screen calculates subtotal, service charge, sales tax, discount, and grand total.
- Choose Dine-in (with table number) or Takeaway (with pickup time) before placing the order.

Order Tracking (Customer)

- Each placed order automatically progresses through Pending → Preparing → Ready → Served.
- A live elapsed-time counter shows how long the order has been in progress.

Table Reservations (Customer)

- Select a party size, date, time slot, and table.
- Only time slots with a table large enough and free at that time are selectable.
- Phone number format is validated before confirming.
- View and cancel existing reservations.

Manager Dashboard (Manager only)

- Orders tab: view all incoming orders and manually update their status.
- Reservations tab: accept or decline customer reservation requests.
- Menu tab: edit item prices and toggle item availability on/off in real time.

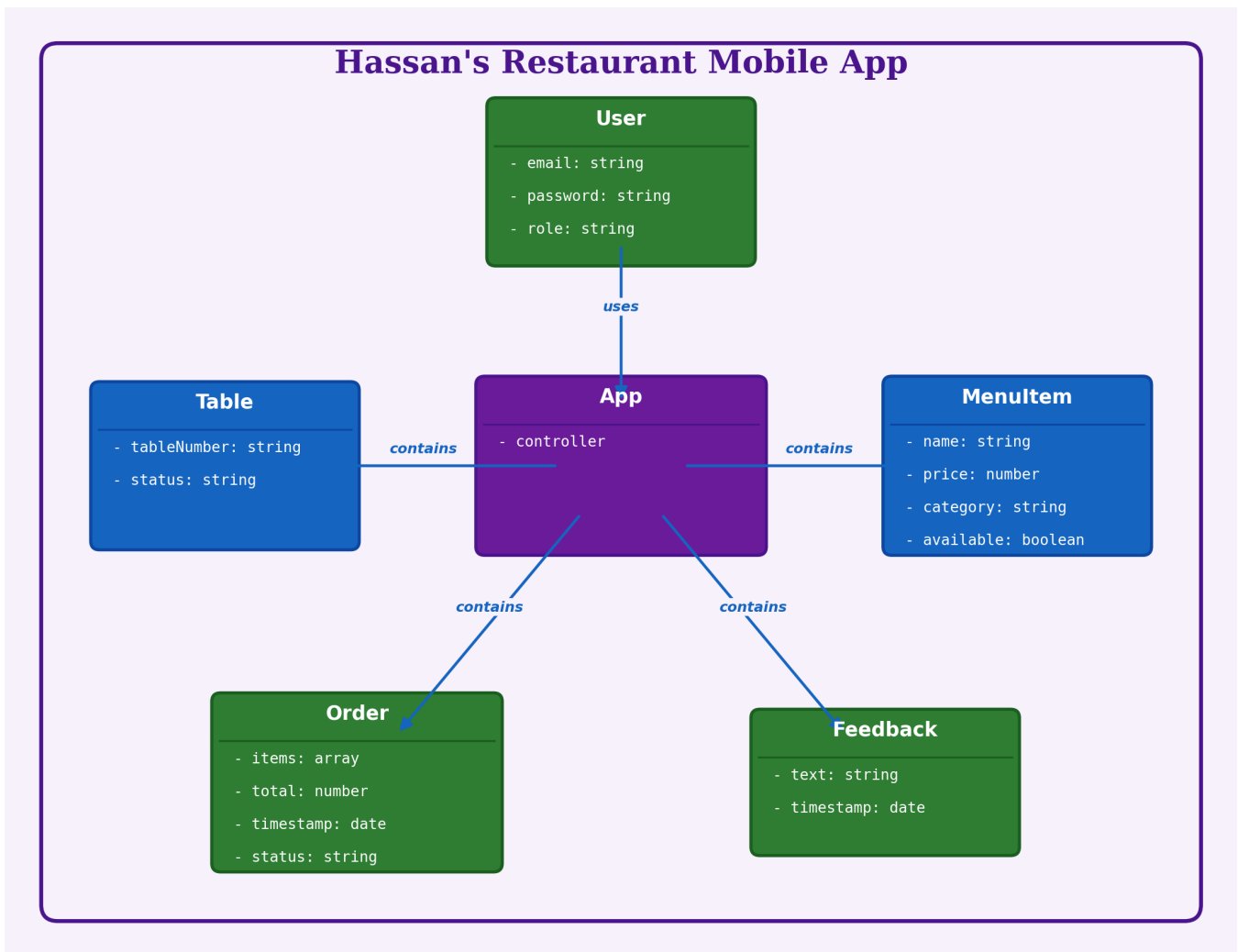
Account & Preferences

- Profile screen shows the logged-in user's name, email, and role.
- Dark mode toggle switches the whole app's color scheme.
- Logout returns to the login screen.

Data Persistence

- Orders, reservations, and manager menu edits are saved to on-device storage (AsyncStorage), so they are still there the next time the app is opened.

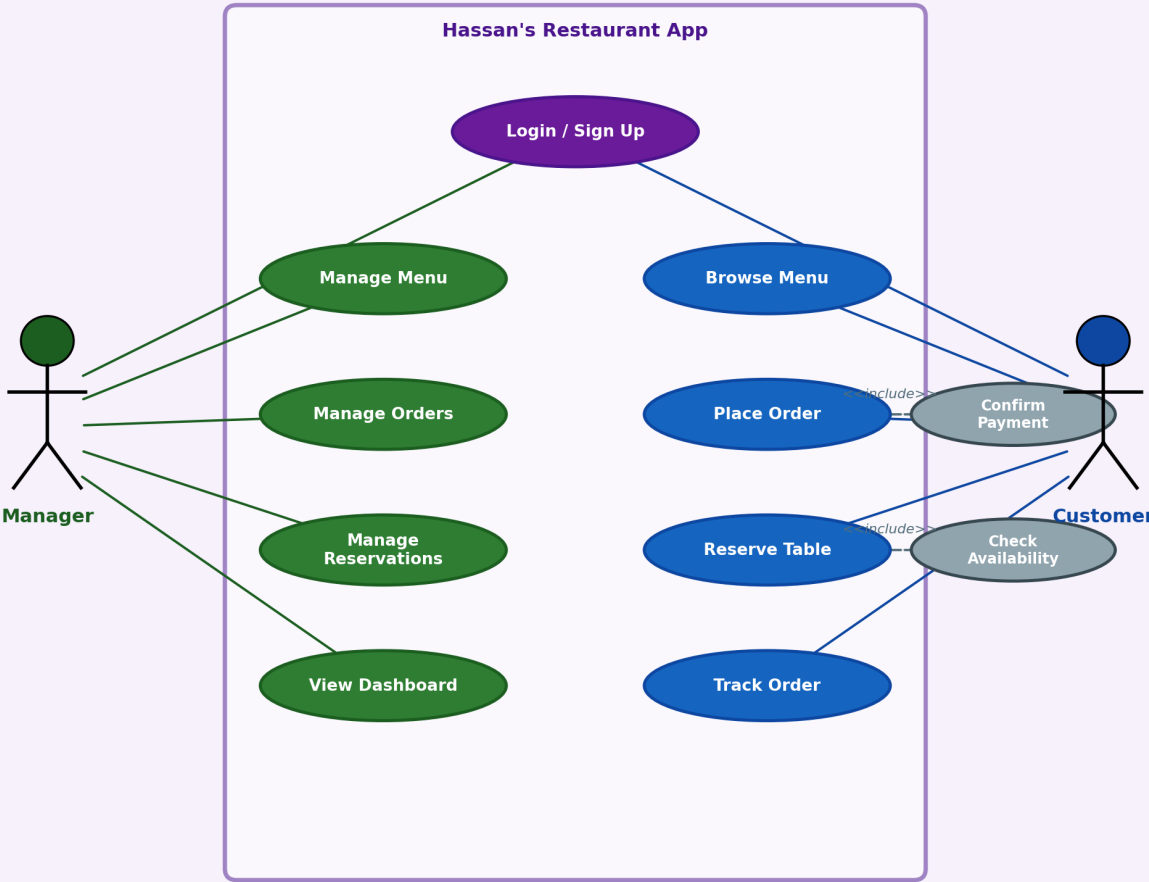
Class Diagram



Use Case Diagram

Hassan's Restaurant Mobile App

Use Case Diagram



Sequence Diagram

